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SURVEY

3D Convolutional Neural Networks for Brain Tumor Analysis in Multimodal MRI: A Systematic Review

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ABSTRACT Convolutional neural networks (CNNs) have emerged as a preferred approach for medical image analysis. The dimensionality of images is a principal factor in CNN models, as they are designed to interpret 2D or 3D image data. This study provides a comprehensive assessment of 3D CNN-based models for brain tumor segmentation and classification of magnetic resonance imaging (MRI) scans, focusing on their performance in terms of accuracy and the Dice Similarity Coefficient (DSC), along with an analysis of self-reported limitations. The integration and processing of diverse data from different MRI sequences, such as resolution, contrast, and noise, pose challenges in the construction of robust models. Consequently, the review focused on multimodal MRIs to provide a comprehensive overview of peer-reviewed literature on 3D CNN models. Therefore, the study performed a systematic literature review (SLR) of articles published between 2019 and 2024 in PubMed, Scopus, and IEEE Xplore databases. The SLR initially identified 554 potentially eligible studies screened for relevance and quality, resulting in the inclusion of 32 studies. Based on these studies, the article conducted systematic and quantitative analysis from technical and task perspectives. The findings indicate that the technical aspects of 3D CNN-based models for brain tumor segmentation and classification can be further improved. The study also discussed the limitations of implementing 3D models for brain tumor analysis. Furthermore, it explores the challenges of translating deep learning (DL) techniques into clinical settings and offers insights into future research trends and advancements.

INDEX TERMS Deep learning, tumor classification, tumor segmentation, 3D convolutional neural network (CNN).

I. INTRODUCTION

In recent years, significant advances in computer vision and deep learning have led to remarkable adaptations in various fields. In the medical domain, rapid advancements have fueled an increasing demand for medical imaging to offer more advanced diagnostic analysis for early and accurate detection of brain tumors and improved long-term survival rates [1], [2]. Magnetic resonance imaging (MRI) remains the preferred imaging modality because of its high sensitivity and non-invasive properties, which are made possible by different scanning sequences [3]. Computer-aided diagnosis (CAD) systems play a vital role in processing and analyzing medical images. In addition, the segmentation and classification of brain tumor tissues using MRI scans are fundamental for follow-up treatment and diagnosis [4]. Remarkable advancements in deep learning (DL) have demonstrated convolutional neural networks (CNN) as a leading method for automatically detecting and classifying brain tumor MRI scans because of their capability to read out hierarchical features directly from images. As the backbone of DL, CNNs have successfully automated complicated tasks, such as segmenting images

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and classification, exceeding machine learning methods [5]. However, the dimensionality of medical imaging data significantly influences the design and performance of CNN models. While most medical imaging datasets are generated from 2D or 3D imaging modalities, many studies have applied 2D CNN models to 3D imaging data, potentially limiting the spatial context capture. Therefore, understanding these aspects is necessary for researchers to successfully use DL models in this field [6]. The main goal of this study is to conduct a comprehensive investigation of 3D CNN-based DL models for the segmentation and classification of brain tumors using multimodal MRI data.

The findings from this study will assist researchers interested in leveraging their DL ability to identify and detect brain tumors. A systematic evaluation of 3D CNN-based models in this context is crucial to determine their efficiency and identify areas for improvement. Researchers can exploit the capabilities of 3D CNN architectures to obtain accurate and productive segmentation and classification by systematically assessing 3D CNN properties, such as their ability to extract volumetric representations using 3D kernels. Recognizing the challenges associated with 3D CNNs, including their exposure to noise, limited availability of training data, high computational demands, and architectural complexity, can inform future research directions and highlight areas requiring further enhancement [7]. This study provides an overview of previous studies, and an analysis of the 3D CNN-based models proposed in the literature between 2019 and 2024, with the aim to introduce a systematic literature review (SLR) of the performance of different 3D CNN-based models in brain tumor analysis using multimodal MRI.

To the best of our knowledge, no comprehensive SLR specifically concentrates on the application of 3D CNNs for brain tumor segmentation and classification using multimodal 3D MRI data. The main objective of this study was to conduct an SLR to bridge this gap by comprehensively analyzing the role of a 3D CNN in brain tumor analysis using multimodal MRI scans. This review highlights how 3D CNNs incorporate spatial and contextual volumetric information from multimodal 3D MRIs and clarifies their strengths and drawbacks compared to other techniques. The results of this review will present valuable insights for researchers in analysing brain tumor through DL approaches. In addition, a comprehensive comparison between various research methodologies will be able to demonstrate the strengths and limitations of the progress in the field and grasp the performance of several DL architectures. More specific objectives can further break this down: (1) to analyze the characteristics of the articles and the implications of using 3D CNN architectures on multisequence MRI data in the detection and/or classification of brain tumors. (2) To discuss the challenges associated with applying 3D DL methods for brain tumor classification and/or segmentation. (3) Discover future research avenues for enhancing the 3D MRI performance of CNN-based models. This SLR aims to address the following research questions: Q1: How do 3D CNN-based models enhance the segmentation and classification of brain tumors when leveraging multimodal MRI data? Q2: How do these models enhance boundary delineation and capture spatial and contextual features by addressing challenges such as intensity inhomogeneity and tumor heterogeneity? Q3: What are the current limitations of 3D CNN-based architecture in the medical imaging field? Q4: What strategies should be explored in the future to overcome these challenges?

II. BRAIN TUMOR

Brain tumors, such as pituitary, gliomas, medulloblastomas, and schwannomas tumors, directly affect brain tissues, including meningiomas. These tumors are categorized based on various characteristics such as growth rate, level of aggression, nature, shape, and origin [8]. The World Health Organization (WHO) categorizes brain tumors into four grades according to their level of aggressiveness. The lowest aggressive tumor was considered a Grade I tumor, exhibiting slow growth. In contrast, Grade IV tumors are extremely aggressive and marked by swift proliferation and widespread tissue destruction. Grade I cells behave like normal cells and grow slowly, whereas grade II cells have some unusual features and grow gradually but can potentially infect the surrounding tissues. Although they are more dangerous than grade I tumors, they are still treatable with proper care. Grade III tumors cause infiltration of healthy tissue, and grade IV cells proliferate rapidly and exhibit destructive behavior [9]. Among gliomas, the most common are high-grade gliomas (HGG) and low-grade gliomas (LGG). According to the WHO, LGG tumors, such as diffuse astrocytomas and oligodendrogliomas, are classified into grades I and II, whereas HGG tumors, which include glioblastomas (GBM), are categorized as grades III and IV [10].

Brain tumors have complex structures with different regions, including the whole tumor (WT), tumor core (TC), and enhancing tumor (ET). These regions play distinct roles in understanding the tumor structure, which enhances the analysis of brain tumors. As illustrated in Fig. 1, necrotic tissue (NCR), enhancing tumors (ET), non-enhancing tumor regions (NET), and edema are some of the pathogenic components that give rise to these areas [11].

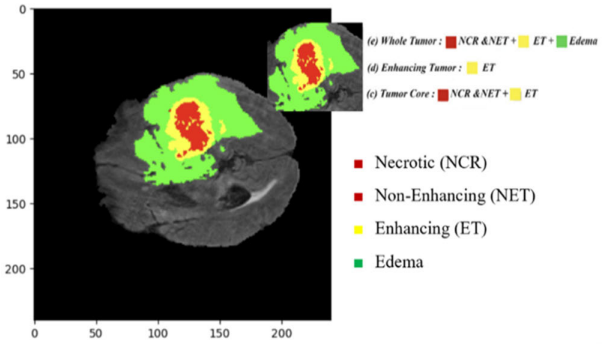


FIGURE 1. Brain tumor regions within the MRI scans.

The WT region presents an entire view of the tumor structure and its impact on the surrounding brain components by encompassing all tumor tissues, including the NCR, NET, ET, and surrounding edema. The TC region highlights the internal components of the tumor, such as the NCR, NET, and ET regions. This region represents the primary part of the tumor for treatment response, as it often includes the most active tissues. The ET region, a component of the tumor core, denotes areas exhibiting hyperintensity on contrast modalities due to the blood-brain barrier [12].

III. MRI MODALITIES

Magnetic resonance imaging (MRI) is a radiation-free imaging approach that produces high-contrast images of organs and soft tissues. The 3D MRI has emerged as an essential tool in diagnosing and treating brain tumors, providing enhanced spatial resolution and volumetric data, especially for evaluating tumor form and development. This facilitates accurate observation of complicated brain structures and tumor profitability, which is crucial for research and clinical applications [8], [13]. Different MRI sequences benefit brain tumor evaluation because each sequence offers distinct tissue contrasts and further details, including T1-weighted (T1), T2-weighted (T2), T1-weighted images post-contrast (T1ce), fluid-attenuated inversion recovery (FLAIR), diffusion-weighted imaging (DWI), and dynamic contrast-enhanced MRI (DCE-MRI) [8]. Table 1 provides an overview of the most widely used types of MRI scans for brain tumor analysis [14].

IV. PERFORMANCE METRICS

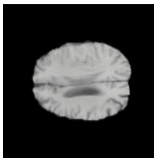
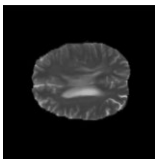
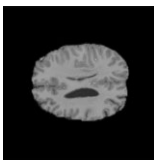
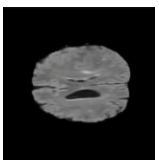
Evaluating the performance of a CNN-based DL model is an essential procedure to provide a comprehensive view of its reliability and effectiveness [8]. The performance metrics help us understand various aspects of the model's performance under changing data conditions. The confusion matrix, a fundamental tool that presents a DL performance vision, is the source of these metrics. In addition, it provides four predictions, including true positive (TP), true negative (TN), false positive (FP), and false negative (FN). The TP presents an image correctly classified into the positive class according to the ground truth. Similarly, the TN is when the model correctly classifies an image into the negative class. On the other hand, the FP is an outcome in which the model incorrectly classifies an image into the positive class when the ground truth is negative. The FN is when the model incorrectly classifies an image that should be placed in the positive class [15]. This study highlights the performance metrics most employed in the literature in Table 2.

V. METHODOLOGY

A. STUDY PROTOCOL

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure a strict and structured approach to systematic review methodology that reflects advances in methods to identify, select, and appraise studies [16], [17], [18], [19].

TABLE 1. MRI modalities characteristics [20].

Type	Feature	Sample
T1	Cerebrospinal fluid (CSF) appears dark, white matter appears bright, and the cortex appears grey. It identifies brain tumor patterns and healthy tissue annotation.	
T2	The CSF shows bright intensity, edema displays very bright intensity, the white matter has dark grey intensity, and the cortex exhibits light grey intensity. It is to detect fluid-filled abnormalities and edema.	
T1ce	It uses a contrast agent to enhance tumor borders, necrotic cores and active cells are clearly distinguishable, and it is more sensitive to detect tumor details better than other sequences.	
FLAIR	Fluid-attenuated to make CSG appear dark, highlights edema and lesions, cortex – Appears light grey, and the white matter appears dark grey. It differentiates edema from CSF and detects brain abnormalities.	

B. ARTICLE SEARCH

Diverse search engines, including Scopus, IEEE Xplore, and PubMed databases, were used between July and September 2024 to identify the relevant literature. Boolean operators and regular expressions were employed to create a comprehensive search query to facilitate the search procedure. After discussion among the authors, a compatible search query with the search engines was as follows: (''deep learning'' OR ''convolutional neural network'' OR ''CNN'' OR ''3D CNN'') AND (''brain tumor'' OR ''glioma'') AND (''classification'' OR ''segmentation'' OR ''detection'') AND (''MRI'' OR ''multimodal'' OR ''multimodal MRI''). At the search stage, a language filter was used to limit the results to only English articles, and a publication year filter was used to restrict the screening process to only papers published between 2019 and 2024. All retrieved studies were imported into Mendeley reference management software. The final selection of articles for review was conducted through a two-stage screening process, guided by predefined eligibility criteria. Initially, duplicate records retrieved from the three databases were removed, followed by screening of the titles and abstracts. In the second stage, articles of the CNN-based model with multiple MRI modalities were checked by reading the full text to confirm eligibility.

TABLE 2. Performance measurements [21].

Metric	Definition	Equation
Accuracy (Acc)	It presents how many correct predictions the model made for the complete test dataset.	$(TP + TN)/(TP + TN + FP + FN)$
Precision (Pre)	It represents the proportion of the correctness of positive predictions out of all the positive predictions.	$TP/(TP + FP)$
Specificity (Spe)	It provides the ability of the model to identify the true negative predictions out of all actual negatives.	$TN/(TN + FP)$
Sensitivity (Sen) / Recall	It provides the ability of the model to identify the true positive predictions among all actual positives.	$TP/(TP + FN)$
F1-score	It provides the average value of precision and recalls metrics to provide a balance between these two metrics.	$2(\text{Pre.Recall}/(\text{Pre} + \text{Recall})$
Dice Similarity Coefficient (DSC)	It identifies the worst-case error in the segmentation by measuring the maximum distance between the predicted boundary points and the ground truth.	$\text{Max}(h(A,B), h(B,A))$

C. INCLUSION/EXCLUSION CRITERIA

Fig. 2 illustrates the article screening process adopted from PRISMA 2020 guidelines [17]. 32 articles were selected for this SLR, focusing on the classification and/or segmentation of multimodal MRI of brain tumors using 3D CNN-based DL models. The study evaluated the performance of 3D CNN architectures to diagnose multisequence brain tumor MRIs using one or more of the following quantitative measurements: accuracy, dice similarity coefficient (DSC), recall, specificity, precision, or F1-score. The following inclusion criteria were used to restrict the scope of this review to the most recent advances in 3D CNN-based DL models:

1) Studies using multimodal MRI data (e.g., T1-weighted, T2-weighted, FLAIR) for brain tumor analysis.
2) Research employing 3D CNN-based models for brain tumor segmentation or classification.
3) Studies reporting quantitative performance metrics such as DSC, recall, accuracy, F1-score, specificity, or precision.
4) Studies focus on either segmentation, classification, or both for brain tumors.

The included studies exclusively focused on the 3D aspects of CNN models employed on multimodal MRI data. Therefore, studies using multiple deep learning techniques and 3D CNN-based models are included in this review, but only the performance of 3D CNN-based models is discussed. To focus exclusively on CNN-based approaches, studies that used traditional machine learning techniques for segmentation or classification were excluded. Studies that used alternative DL architectures, such as generative adversarial networks (GANs) [22], vision transformers [23], recurrent neural networks (RNNs) [3], or capsule networks (CapsNets) instead of CNN models [24], were also excluded. The following exclusion criteria were applied during the screening process to ensure the relevance of the search results:

1) Studies using imaging modalities other than MRI (e.g., CT, PET, or ultrasound).
2) Research employing 2D CNNs only, traditional machine learning methods, or non-deep learning models.
3) Studies focusing on non-3D CNN architecture without direct relevance to 3D CNNs.
4) Studies analyzing non-brain tumor pathologies (e.g., Alzheimer's disease and stroke) or general brain disorders.
5) Studies either did not report quantitative performance metrics or lacked an evaluation of the proposed method.
6) Review papers, meta-analyses, or theoretical works without experimental results.

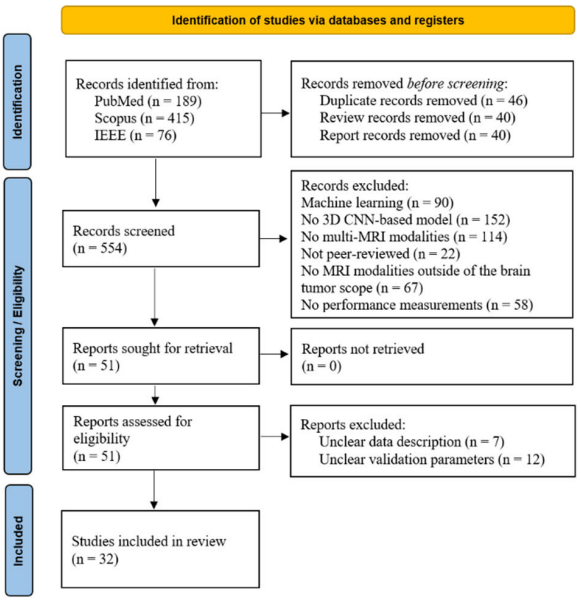


FIGURE 2. PRISMA 2020 flowchart of the article screening process; illustrates the selection of this SLR (Adopted from Page et al. [17]).

D. QUALITY ASSESSMENT

To ensure the quality of the content of each study, two reviewers independently screened all retrieved studies at the title, abstract, and full-text levels based on the predefined eligibility criteria. Shaffril et al. [18] recommended that experts classify the studies into one of three quality categories: high quality, medium quality, or low quality. Only articles rated as high or medium quality should be considered for evaluation. Therefore, the following quality criteria were used to evaluate

the included studies: (1) Is the study related to the research objectives? (2) Does the study relevant to the 3D CNN models for brain tumor diagnosis? (3) Is the study content understandable and clear? (4) Is the study methodology and the dataset clearly explained?

VI. RESULTS

CNN models have revolutionized feature engineering by leveraging their self-learning capabilities. Gradually, CNNs architectures have been developed and have exhibited remarkable results in brain tumor analysis over time [25]. The scope of this study was to analyze how 3D CNNs have been applied to brain tumors, evaluate the performance of these models, and identify gaps and directions in the brain tumor analysis field by critically reviewing 32 studies. Following the PRISMA guidelines [17], the SLR was conducted in several stages to meet its objectives [26]. Initially, the SLR commenced with a search for relevant studies according to a query constructed based on a set of keywords in the literature. As illustrated in Fig. 2, the search retrieved 680 publications between 2019 and 2024 from three databases: 415 publications from Scopus, 189 publications from PubMed, and 76 publications from IEEE Xplore. After removing duplications, reviews, and reports, 554 studies remained for title and abstract screenings. Articles not aligned with research goals were excluded from consideration at this level. A total of 51 articles were screened by reading the full text with additional criteria to meet the eligibility criteria. As indicated in Fig. 2, 32 publications were selected as the final included studies in this SLR because they met all the eligibility criteria while reading the full text. The 32 included studies differed in the tasks they sought to address, such as segmentation, classification, and hybrid frameworks, as shown in Fig. 3.

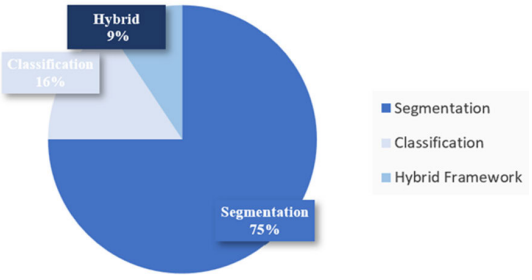


FIGURE 3. Percentage of the states of research by the framework.

A. MULTIMODAL MRI DATASETS

Training datasets play a crucial role in the learning process of DL models through feature extraction from the patterns inherent in the dataset, which is critical for the performance model [27]. Multimodal MRIs, including T1, T1ce, T2, and FLAIR sequences, are essential for providing complementary information on tumor behavior from various planes. This diversity of MRI modalities contributes to the identification and detection of tumor region boundaries such as WT, TC, and ET [28]. The field of brain tumors contains several 3D MRI datasets featuring multimodal brain structures such as BRATS [29], [30], TCGA-LGG [31], TCGA-GBM [32], REMBRANDT [33], CPM-RadPath [34], and ISLES'22 [35]. The BRATS dataset, available in multiple versions, was the main data source for most of the included studies. It is considered a cornerstone for analyzing 3D CNN-based models for brain tumor segmentation and classification since it offers the most common 3D MRI modalities (T1, T1ce, T2, and FLAIR) and ground truth for segmentation. However, despite its widespread use, this dataset poses notable limitations. First, the BRATS versions suffer from small sample sizes (e.g., ∼300–400 cases per BRATS version), which constrains the capacity of DL models to generalize effectively to unseen data [36]. Second, inter-observer variability in manual tumor annotations introduces annotation noise and inconsistencies, potentially affecting segmentation quality [37]. Third, specific datasets may contain institutional biases, limiting the clinical applicability of trained models across diverse patient populations [38]. Several studies have incorporated data augmentation techniques such as rotation, flipping, scaling, and elastic deformation to increase data diversity and reduce overfitting synthetically to address these challenges [39]. The following data preprocessing section shows the use of data augmentation techniques and demonstrates their role in improving models' performance in the included studies.

Furthermore, transfer learning has emerged as a robust solution to the data scarcity problem, allowing models pre-trained on large-scale datasets (e.g., ImageNet) to be fine-tuned on small medical datasets with limited samples. In other words, transfer learning involves taking what the network has learned through training on an extensive dataset (the base dataset) and applying it to a smaller dataset (the target dataset) [40]. Despite this strategy improving the model's robustness by transferring learned representations, a critical need remains for sharing annotated 3D multimodal MRI datasets since most of the pre-trained models are structured in a 2D manner [41]. Table 3 outlines each version of the BRATS dataset, detailing the cases in each set, classes, annotations, and related articles. In contrast, two notable studies [42], [43] employed their models on a combination of 2D and 3D MRI datasets. Rani et al. [42] used the Figshare dataset [44], which consists of 3064 2D images of 233 patients, a medical dataset collected from the Pentagram Research Institute in Hyderabad, and the Kaggle dataset [45], which comprises 253 2D MRI scans, along with BRATS 2019. In this study, horizontal rotation was employed to create multiple representations of a 3D model. Zhuge et al. [43] selected 30 cases from TCGA-LGG [31] and 285 patients with 3D MRIs from BRATS 2018 to apply their brain tumor segmentation and classification models.

B. DATA PREPROCESSING

Various standard preprocessing techniques have been employed to enhance the quality of data input and improve the model training process. Table 4 shows the studies that

used these preprocessing techniques in their works, such as bias correction [46], normalization [47], [48], denoising [49], cropping [50], resizing [51], contrast enhancement [52], and data augmentation [39].

The cropping approach removes the background around the brain region to focus on this specific region, whereas resizing changes the image dimensions by reducing or increasing their size [51]. In two studies, Ilyas et al. [56] and Wu et al. [58] used cropping as the main preprocessing technique by standardizing foreground tumor region slices and removing the background proportion to obtain cropped tumor patches. In Zia et al. [64], the authors performed the cropping technique on T1, T2, and FLAIR scans to obtain a size of 128 × 128 × 112 because of their large sizes. Similarly, Aboussaleh et al. [77] resized 3D images to 128 × 128 × 128 pixels but resizing techniques were performed by ignoring the first 15 and last 12 slices.

Normalization is required to improve the performance of the CNN and ensure that the input data of the model has similar intensity distributions [84]. In the literature, two main normalization approaches are widely used to normalize the input data: z-score and min-max normalization. The z-score normalization refines data consistency by subtracting and dividing the mean value by the standard deviation [84]. For example, Ali et al. [73] employed z-score normalization on whole volumes of training and validation datasets, including four modalities and corresponding segmentation mask, to eliminate intensity contrast and easily identify the tumor region. Additionally, Li et al. [62] incorporated z-score normalization and cropping and then chunked the 3D image into more manageable pieces with sizes of 32 × 160 × 160 to overcome the limitation of computational efficiency. In contrast, min-max normalization scales the data to (0, 1) by subtracting the minimum value and dividing it by the range of values [47]. For instance, Montaha et al. [60] employed min-max normalization on all 3D images to obtain a standard scale of (0-1) and then resized all images to 128 × 128 × 32 to eliminate computational complexity. Mzoughi et al. [61] applied min-max normalization to the T1ce modality to rescale the intensity; then, they developed adaptive contrast stretching based on statistical information from the scan. They evaluated model performance with and without preprocessing techniques. In three other studies, Aboussaleh et al. [77], Khan et al. [79], and Rutoh et al. [78] measured the mean and standard deviation attributes of 3D small patches with dimensions 128 × 128 × 128 and then concatenated the four modalities to form a 128 × 128 × 128 × 4 stack.

To solve the intensity nonuniformity problem caused by the MR scanner or patient movement, Yin et al. [72] applied N4 insight segmentation and registration toolkit (N4ITK) correction using a Gaussian filter and applied the B-spline of the intensity to adjust the brightness and generalize the data. This led to a more consist of intensity profile across MRI scans by reducing artifacts introduced by scanners. Similarly, Ullah, et al. [69] combined the N4ITK bias correction with Nyul intensity and z-score normalization to improve input heterogeneous data quality [85]. Adaptive histogram equalization (AHE) was then employed to enhance contrast and Gibbs ringing artifact removal to sharpen the tumor boundaries.

By contrast, Rani et al. [42] processed all the 3D MRI scans to remove noise from each scan using an adaptive median filter. Their technique compares each pixel with its neighborhood and replaces any different pixels identified as noise. Meanwhile, Zaitoon and Syed [57] applied a convolutional normalized mean filter (CNMF), which combines convolutional and normalized mean filters to smooth the image by replacing noise pixels with convolved median values that can ensure that the image information remains intact. In the study by Dang et al. [76], gamma correction was employed to enhance MRI intensity with gamma values applied to three MRI modalities (T1, T1ce, and T2). The authors also performed a window-setting optimization (WSO) approach with a customized linear function.

Data augmentation is a strategy to increase data diversity for the training process of CNN-based DL models to address the lack of data availability in medical imaging. This strategy is accomplished by employing different manipulation techniques on the original data or by generating new samples through generative models [39]. The augmentation techniques have been applied to brain tumor analysis, including rotation [43], [55], [58], [63], [66], [73], [74], [75], [76], flipping [43], [55], [61], [63], [65], [70], [71], [75], [80], [83], shifting [70], [83], translation [58], [81], clipping [58], [65], cropping [56], [75], deformation [83], scaling [43], [55], [83], and Gaussian noise [71], [73], [81], [83].

Magadza and Viriri [81] randomly applied multiple augmentation approaches such as Gaussian noise, scaling flipping, and translation, with a 50% probability for each training iteration. Wang et al. [70] augmented the training data (−0.1 to 0.1) variance shift and (0.9 to 1.1) intensity shift with 0.2 probability. Subsequently, Mzoughi et al. [61] demonstrated that the flipping approach is a significant augmentation technique compared with other strategies for medical images. Fig. 4 illustrates the publications from 2019 to 2024 related to the preprocessing techniques used in the included studies. Notably, the total number of publications peaked in 2022, whereas other techniques showed varying trends over the years.

C. CLASSIFICATION

Classification is a fundamental task that predicts and sorts of data points into predefined groups called classes. By training models on labeled data, classification algorithms learn to observe different classes and make accurate predictions for new, unseen data [1]. Recently, CNNs have emerged as dominant tools for image classification because of their ability to extract hierarchical features from data using feed-forward connections between the layers [86]. Various effective CNN

TABLE 3. The characteristics of the annual BRATS datasets [53].

Dataset Name	Training Set	Validation Set	Test Set	Classes	Annotations	List of References
BRATS 2012 - 2013	80	N/A	N/A	N/A	Four tumor classes manually annotated by experts (NCR, NET, ET, and Edema)	N/A
BRATS 2014 - 2016	274	N/A	BRATS 14: 38, 15: 53, and 16: 191	220 HGG and 54 LGG cases.	Same as BRATS 2012 - 2013.	N/A
BRATS 2017	285	46	146	210 HGG and 75 LGG cases.	BRATS 2012 – 2013 + 19 institute MRI scans.	[54], [55], [56], and [57].
BRATS 2018	285	66	191	210 HGG and 75 LGG cases.	BRATS 2012 – 2013 + 19 institute MRI scans.	[43], [55], [56], [57], [58], [59], [60], [61], [62], [63], [64], [65], [66], [67], [68], [69], [70], and [71].
BRATS 2019	335	125	191	259 HGG and 76 LGG cases.	BRATS 2012 – 2013 + 19 institute MRI scans.	[42], [57], [58], [60], [62], [63], [64], [66], [68], [72], [73], [74], [75], and [76].
BRATS 2020	369	125	191	293 HGG and 76 LGG cases.	BRATS 2012 – 2013 + 19 institute MRI scans.	[57], [60], [63], [64], [65], [68], [70], [77], [78], [79], [80], [81], and [82].
BRATS 2021	1251	125	191	MGMT	BRATS 2012 – 2013 + 19 institute MRI scans.	[83]

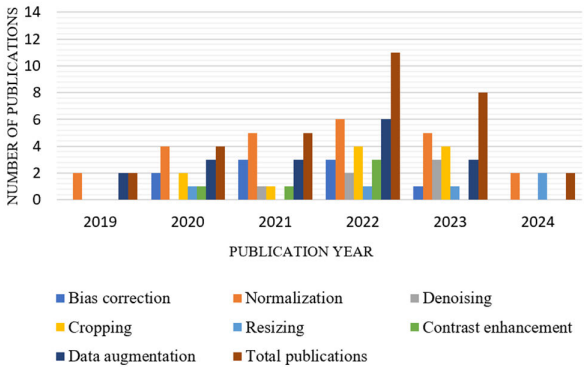


FIGURE 4. Utilization of preprocessing techniques for 3D MRI from 2019 to 2024.

models have been established to address the classification and segmentation tasks. CNNs consist of different layers that are stacked sequentially, including convolution layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification [87]. Because CNN models are constructed to analyze image data, the dimensionality of images plays a crucial role in model development. Most medical image data are generated from 2D or 3D imaging modalities. However, according to Niyas et al. [6], most published studies have analyzed 3D data using 2D CNN models. The 3D CNN network structure resembles the standard 2D CNN model, as depicted in Fig. 5, while the convolution and pooling layers use 3D extensions to process the volumetric data [6], [7]. The primary function of 3D CNNs is to extract spatial and contextual links within volumetric data using a kernel in three spatial dimensions (x, y, and z), which allows the model to capture inter-slice information from multiple views to obtain hierarchical features [88].

Rani et al. [42] and Mzoughi et al. [61] proposed 3D CNN-based DL architectures for glioma grading using 3D MRI data from the BRATS datasets. Mzoughi et al. [61] presented a more robust approach than the study by Rani et al. [42]. They introduced a different deep multi-scale 3D CNN model with preprocessing steps, including normalization and adaptive contrast enhancement, to address intensity heterogeneity in data. Then, leveraged the model with eight small convolutional layers ($3 \times 3 \times 3$) to extract global and local features that can decrease the challenges, such as model complexity and computational cost. Their model evaluated 284 cases and achieved promising results with a 96.49% accuracy rate for glioma grading. In contrast, Rani et al. [42] deployed a 3D CNN-based approach using the 3D AlexNet model and a Cellular Logic Array Processing (CLAP) algorithm to extract wireframe representations of tumor structures. The basic patterns within the 3D images were extracted by scanning the entire scan to structure a 256-dimensional vector of features for classification, which resulted in an accuracy of 96.91%. Mzoughi et al.'s [61] model is superior due to its optimized 3D CNN design, preprocessing, simplicity, and better accuracy, making it a more scalable solution for the classification of MRI brain tumor. Mzoughi et al.'s [61] model is superior due to its optimized 3D CNN design, preprocessing, simplicity, and better accuracy, making it a

TABLE 4. List of preprocessing techniques used by the included studies.

Preprocessing Technique	Studies Using It	Performance Impact	Key Challenges	References
Z-score Normalization	16/32 (50%)	Improved DSC by 3-7% and enhanced generalization of the model across various MRIs.	Sensitive to outliers, it may not reduce inter-scanner variability.	[43], [55], [62], [63],[69], [70], [71], [72], [73], [74], [75], [77], [78], [79], [80], and [82].
Min-Max Normalization	5/32 (16%)	Enhanced contrast facilitated faster convergence during training.	It may compress intensity ranges, potentially losing subtle details.	[60], [61], [64], [65], and [66].
Cropping	12/32 (37%)	It focuses on the region of interest (ROI) and reduces irrelevant background noise.	It may eliminate relevant surrounding tumor tissues if not performed carefully.	[56], [58], [60], [62], [64], [65], [67], [68], [70], [72], [75], and [82].
Resizing	5/32 (16%)	It reduces computational cost by standardizing input size.	Potentially lose relevant details and introduce artifacts.	[60], [61], [77], [78], and [79].
N4 Bias Field Correction	8/32 (25%)	Improving segmentation accuracy by correcting the inhomogeneities.	It requires careful parameter tuning.	[43], [54], [66], [67], [69], [70], [72], and [83].
Denoising	5/32 (16%)	It minimizes noise and artifacts, making tumor borders easier to distinguish and more robust across datasets, and reducing computing complexity.	Over-smoothing may remove subtle tumor characteristics, reducing sensitivity for small or complex tumors.	[42], [57], [59], [69], and [80].
Contrast Enhancement	5/32 (16%)	Improving the visual quality of images by making details more distinguishable.	It can be computationally intensive, impacting processing time in real-time applications.	[42], [59], [61], [66], and [69].
Data Augmentation	18/32 (56%)	The process involves mitigating class imbalance and dataset biases, preventing overfitting, and enhancing data variability.	Some augmentation techniques require high computational cost, increasing training time.	[43], [55], [56], [58], [61], [63], [65], [66], [68], [70], [71], [73], [74], [75], [76], [80], [81], and [83].

more scalable solution for the classification of MRI brain tumor.

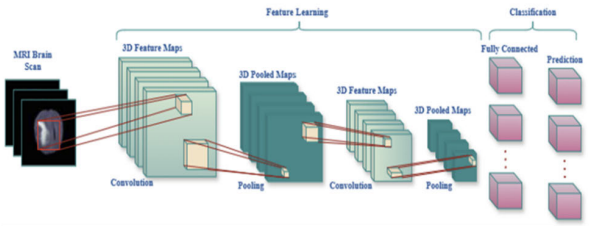


FIGURE 5. Illustration of 3D CNN structure for brain tumor analysis [7].

Wu et al. [58], Molaei et al. [59] and Montaha et al. [60] introduced hybrid and attention-based models to improve the classification performance further. Montaha et al. [60] proposed a hybrid time-distributed CNN-LSTM (TD-CNN-LSTM) that integrates a 3D CNN with long short-term memory (LSTM), which is superior to the others due to its advanced architectural design and practical strengths in feature extraction. The TD method was applied to each layer of the 3D CNN to process all four MRI modalities for each patient as a single input data owing to the necessity of tumor information in every sequence. Therefore, this approach enables simultaneous spatial feature extraction from volumetric data and temporal dependency modeling using LSTM. Their ablation study on BRATS 2020 achieved a peak test accuracy of 98.9% and optimized GPU resource constraints with a total model complexity of 116 million operations, outperforming the standalone 3D CNN model. In contrast, the study by Wu et al. [58] incorporated the attention-based glioma grading network (AGGN) by a dual-domain attention mechanism, multi-scale feature extraction, and multimodal fusion for glioma grading. These techniques allow the AGGN to highlight critical regions and features across four MRI modalities, leading to better discriminative power without manually labelling tumor masks; however, it relies heavily on augmented data and is limited in computational demand due to the model complexity.

Meanwhile, Molaei et al. [59] presented a new fuzzy deep convolutional neural network (FDCNet) incorporating fuzzy logic and fractal-based morphological operations. The model was validated on 145 patients from the BRATS datasets and achieved a high performance with an accuracy of 98.68%. However, its segmentation relies on handcrafted feature extraction methods and lacks the adaptive learning dynamics of AGGN.

D. SEGMENTATION

Segmentation is the most popular task in brain tumor diagnosis to accurately delineate tumor boundaries by assessing each patch extracted around each pixel and feeding them to the classifier [4]. The CNN-based segmentation models followed the encoder-decoder architecture of SegNet [89]. The U-Net architecture, a CNN-based encoder-decoder proposed in 2015, comprises shrinking and expanding paths. This model has recently become the cornerstone for segmenting brain tumor applications. The architectural strength of this model stems from the integrating up-sampling and down-sampling layers through skip connections, which establish direct links between the corresponding convolutional and deconvolutional layers, as shown in Fig. 6 [7], [90]. Subsequently, enhanced U-Net models have been proposed to segment 3D volume-to-volume with remarkable performance [91]. The 3D CNN-based models can evaluate intensity patterns across axial, sagittal, and coronal planes, which improve segmentation performance where tumor borders are inconsistent and difficult to delineate [92]. In a study by Shaukat et al. [82], the 3D U-Net model was adapted to a cloud-based architecture optimized for voxel-wise segmentation by leveraging random patch inputs and efficient convolutional layers for scalability. Each encoder module extracts features using two convolutional layers trailed by a ReLu layer to provide non-linearity and ends with a down-sampling layer. The first three decoders up-sample the image to double in the decoder module. Their study achieved an average DSC of 95% across a test set. Similarly, Li et al. [62] proposed a modified 3D U-Net to address the limitations of 2D architecture, particularly in capturing the spatial relations between volumetric information. They redesigned the encoder and decoder structures to capture high-level semantic information while retraining spatial details. They also employed a hybrid loss function with binary cross-entropy to address the class imbalance problem. The results of their model on the WT region were constrained, where the DSC of WT was 77.8%, whereas 87.5 for TC and 90.3% for ET regions, because of information loss in block-based processing. In Ullah [69], a 3D U-Net was enhanced to evaluate the performance of preprocessing techniques, such as bias field correction, Gibbs ringing artifact removal, and intensity normalization. Based on a technical comparison of the three studies, Li et al. [62] offered the most technically advanced and well-justified approach, particularly in the context of brain tumor segmentation using 3D U-Net architectures. Unlike Shaukat et al. [82], whose novelty lies more in cloud-based deployment than in architectural innovation, or Ullah et al. [69], which primarily focuses on pre-processing enhancements (e.g., Gibbs artifact removal and bias field correction).

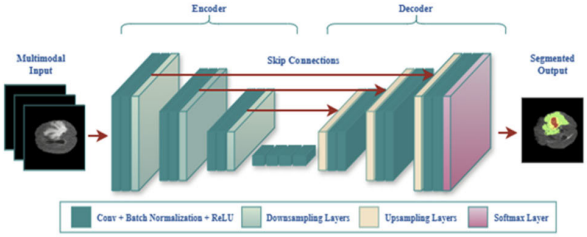


FIGURE 6. An example of brain tumor segmentation using 3D U-Net [7].

Nazir et al. [80], Ali et al. [73], and Nehru and Prabhu [66] have focused on a new convolutional architecture to optimize the performance of the segmentation. Nazir et al. [80] developed a more robust advanced approach by a 3D Kronecker convolution feature pyramid network (KCFP) [93], which consisted of a 3D feature selection (3D FSC) network for extracting detailed lesion features, a feature pyramid for learning features at different scales, and a 3D connected component analysis (3D CCA) step to eliminate false positive values. The KCFP model improves segmentation by exploiting inter- and intra-dilation factors to avoid the pixel-loss problem in standard convolutions while preserving local and global context across pyramid skip connections. Their model allows for effective segmentation of tumors of varying sizes with minimal parameter increase, outperforming others with DSC results for the WT, ET, and TC regions of 0.90, 0.80, and 0.80, respectively. In an identical study by Ali et al. [73], the authors employed a multilevel Kronecker CNN (ML-KCNN). However, it lacks the integrated multi-scale fusion structure and has less architectural innovation beyond post-processing with conditional random fields (CRF) and CCA. Despite its achieved competitive performance (DSC of 0.90 WT, 0.83 TC, 0.74 ET), it does not incorporate advanced feature reuse or pyramid-level fusion mechanisms. In contrast, Nehru and Prabhu [66] presented a depth-wise separable dense U-NET (DSDU-Net) designed to address brain tumor segmentation challenges and capture spatial features while reducing computational demands. It demonstrates efficiency regarding resource usage, but it falls short in terms of advanced contextual modeling seen in the pyramid-based models. Technically, Nazir et al. [80] presented the most efficient segmentation architecture due to its multi-scale Kronecker pyramid structure and balanced performance across tumor subregions.

Two studies applied the attention mechanism technique to assess the robustness and computational efficiency of the segmentation tasks. Khan et al. [79] developed a hybrid attention-based residual U-Net (HA-RUnet) that integrates 3D residual blocks, squeeze excitation (SE), and attention modules. The first step in their framework was to establish a 3D residual U-Net. Multi-scale features are combined with high-level features using different stacked

residual blocks. The encoder-decoder is linked by the skip connections along various attention modules to learn the different attention-aware features of different tumor regions, and the SE enhances these crucial features. Their model outperformed the standard Residual U-Net and ResUNet segmentation precision and efficiency, achieving DSC values of 0.867, 0.813, and 0.787 in the WT, TC, and ET regions, respectively. However, using resized 3D MRI leads to potential information loss due to computational resource limitations. In the second study, Zhou et al. [67] introduced a tri-attention fusion network that combined modality, spatial, and correlation attention mechanisms within a multi-encoder 3D U-Net model. Their model captures inter-modal dependencies by modelling feature correlations using a correlation attention module constrained by Kullback–Leibler divergence [94]. It improved performance compared to the baseline by achieving a DSC average of 0.811. Overall, the tri-attention fusion strategy and correlation-aware learning in Zhou et al. [67] offer a more comprehensive and modality-sensitive segmentation than Khan et al. [79].

Several studies have presented hybrid and ensemble architectures to address the challenges of class imbalance and the segmentation of small tumor parts. Accordingly, Liu and Cheng [83] introduced an enhanced 3D brain channel space attention (3D BCS_T) model to deal with class imbalance and better spatial feature learning. Their study integrated a 3D hybrid attention mechanism and residual blocks. The encoder uses a convolution kernel to combine spatial features, which are then sent to the transformer module for global context modeling. Experiments were conducted in a consistent experimental environment to validate the effectiveness of the hybrid attention mechanisms and residual blocks within the overall model. Despite their model outperforming other methods, such as TransBTS and the convolutional block attention module (CBAM U-Net), with an average DSC progress of 2.53%, the depth of the network architecture led to an increase in the parameters, thus demanding high complexity and computation. In contrast, Ilyas et al. [56] proposed a more robust advanced solution for addressing class imbalances and improving small-tumor segmentation. The authors created a multi-dilated attention network (Hybrid-DANet) for brain tumors. They incorporated several novel components into their model: (1) a hybrid weight alignment with a multi-dilated attention module (HWADA) to extract contextual features, (2) a residual module (RM) to decrease the vanishing gradient problem, and (3) a multi-channel multi-scale module (MCS) to capture different geometric and spatial features across tumor regions. The architecture processes MRI images as concatenated 3D volumes by the skip connections and attention gates. The DSC values for the WT, TC, and ET were 0.892, 0.764, and 0.680, respectively. As a decisive point, Ilyas et al. [56] provide a superior solution through architectural innovation, multi-module unit, and inclusive evaluation, making it more suitable for high-precision and computationally efficient brain tumor segmentation in multi-modal MRIs.

In 2024, Aboussaleh et al. [77] created 3DUV-NetR+, a hybrid encoder-decoder model that combines 3D U-Net and V-Net architectures with transformer modules, to segment brain tumors in multimodal MRI data. Their model used the encoder's ability to extract features based on depth and combined them from 3D U-Net and V-Net to improve spatial representation. On the other hand, Ali et al. [75] introduced an ensemble framework combining two CNN-based models, including a 3D CNN and a 3D U-Net. The 3D CNN outperformed U-Net in ET segmentation, whereas U-Net excels in TC delineation. For WT segmentation, the contributions from both models were equally effective. While Ali et al. [75] achieved competitive DSC (e.g., 0.906 for WT), Aboussaleh et al.'s model outperformed with higher segmentation accuracy (e.g., 0.919 for WT, 0.828 for TC, and 0.817 for ET). In addition, Aboussaleh et al. [77] assessed their model on the more recent BRATS 2020 dataset. They comprised transformer-based self-attention mechanisms, giving it a significant technical edge in learning spatial hierarchies across volumetric data. Therefore, from a deep learning perspective, 3DUV-NetR+ offers a more sophisticated and robust solution to brain tumor segmentation tasks.

Computationally efficient models have been a focus of literature. Magadza and Viriri [81] suggested a modified 3D U-Net with partial depth-wise. This technique reduces the model size and parameters while maintaining competitive segmentation accuracy. Compared to the baseline models, the architecture reduced the model size and parameter count by ∼90% and ∼70%, respectively. The network attained DSC scores of 87.3%, 85%, and 76.6% for the WT, TC, and ET, respectively. However, the model demonstrated variability in sensitivity for ETs and relied heavily on small input patches to improve segmentation metrics. By incorporating dense blocks into the encoder and residual blocks into the decoder, Tie et al. [74] introduced a modified 3D U-Net architecture to enhance the feature propagation and mitigate gradient vanishing problems. Dense blocks, characterized by layer interconnections, are employed in the encoder to strengthen the feature reuse and reduce the parameters. In contrast, residual blocks in the decoder aim to address the performance degradation with deeper layers, ensuring efficient information flow. While their approach yields solid DSC (0.901 for WT, 0.815 for TC, 0.766 for ET on BRATS 2019), it is more computationally demanding, using a GTX 1070 GPU with limited information on model size.

Other studies have employed innovative architectural U-Net-based designs. For instance, a new multi-threading dilated convolution network (MTDC-Net) was proposed by Chen et al. [65] for the automatic segmentation of brain tumors. To learn low-dimensional features in the encoder, they suggested the MTDC approach and spatial pyramid convolution (SPC) to utilize the contextual semantic information in the decoder. Multi-threading adaptive pooling up sampling (MTAU) was applied to increase the resolution of the feature maps. This study was improved by integrating the contextual and structural features within the scales.

Experiments were conducted on BRATS 2018 and 2020 to evaluate their strategies, and the DSC rates were 0.809 (ET), 0.892 (WT), and 0.832 (TC). A new architecture has been proposed by Zia et al. [64] called context-aware 3D ARDUNet architecture. Residual dropout blocks (RDBs) and attentional residual dropout blocks (ARDBs) are the main components of context-aware 3D ARDUNets. Their model was trained on multiple datasets, including BRATS 2018, 2019, and 2020, which achieved a superior DSC of 0.9020–0.9275 for WT and a high sensitivity of up to 0.9843, outperforming baseline methods. Yin et al. [72] designed a novel model by integrating ConvNeXt blocks into the encoder path, called the WD-UNeXt model, to enhance the U-Net architecture and improve the segmentation of brain tumors. The model follows a standard encoder-decoder approach with a bridge for feature refinement, whereas dropout regularization and weighted loss function address overfitting and class imbalance, respectively. The model achieved a competitive DSC of 0.934 for WT, 0.911 for TC, and 0.851 for ET. While MTDC-Net is computationally efficient, ARDUNet offers broader generalization and fine-grained contextual segmentation. WD-UNeXt excels in feature representation and regularization, making each suitable under different technical constraints.

In this literature, different studies have concentrated on multi-stage or cascading frameworks to enhance segmentation outcomes. Lachinov et al. [71] developed a cascaded 3D U-Net architecture for brain tumor segmentation. In this study, the model used cascading U-Net classifiers with iterative refinement to improve segmentation results. Each U-Net model individually processes T1, T2, T1ce, and FLAIR modalities with various encoders to learn the features before combining them. The model produced competitive Dice scores of 0.878 for WT, 0.72 for ET, and 0.795 for TC on the BRATS 2018 test set. However, iterative refinement complicates the design, and inconsistent class distributions and data-specific preprocessing procedures restrict its application. Zhou et al. [54] surpass this approach by introducing a three-stage network that strategically fuses multimodal MRI data using both channel and spatial attention modules. The first stage used a 3D U-Net architecture to generate predictions, whereas the second used a multi-encoder network to retrieve modality-specific features using an innovative attention mechanism. Subsequently, two encoder stages of the 3D U-Net were used to integrate the predictions with raw MRI modalities. The model achieved a DSC of 89.4%, 81.6%, and 73.0% for WT, TC, and ET, respectively. Their study achieved competitive performance and maintained lower memory and computational demands. Therefore, from a technical standpoint, Zhou et al. [54] presented a more innovative and resource-efficient solution for multimodal MRI of brain tumors.

Eventually, multi-scale and hierarchical models were applied to address the segmentation process. Wang et al. [55] proposed a triple-cascaded framework utilizing anisotropic 2.5D CNNs to segment hierarchical brain tumor subregions (WT, TC, and ET) from multimodal MRIs. The framework is efficient in memory and introduces uncertainty estimation. However, it sacrifices full 3D spatial context and relies on multiple networks for hierarchical area prediction, which adds complexity to the model. In addition, error propagation throughout the stages was challenging despite the cascaded framework obtaining a high DSC (0.902 WT, 0.858 TC, 0.797 ET) in BRATS 2018. Meanwhile, Rutoh et al. [78] proposed a guided attention inception residual U-Net (GAID-U-Net) for brain tumor segmentation. This study developed a traditional U-Net using a guided attention mechanism, an inception module, and residual connections to extract hierarchical features from 3D volumetric data. The performance of the GAID-U-Net model was evaluated on the BRATS 2020 dataset, which achieved DSC scores of 0.8870, 0.8621, and 0.8573 for the WT, TC, and ET, respectively. However, their model lacks explicit parameter efficiency analysis and generalization across datasets. In contrast, the multi-scale hierarchical U-Net (MH-UNet) architecture by Ahmad et al. [68] can be considered technically superior because of its architectural balance between parameter efficiency by applying multi-scale strategies, dense connections, and residual inception blocks. The model demonstrated scalability by achieving state-of-the-art results on the BRATS 2018-2020 and ISLES datasets while being 14× lighter than the best-performing model in BRATS 2018 and using 2.2× fewer parameters than the top 3D architectures on ISLES.

Wang et al. [70] and Zhang et al. [63] both proposed lightweight architectures for brain tumor segmentation. Wang et al. [70] provided a lightweight hierarchical convolutional network (LHC-Net), which aimed to reduce the computational requirements without sacrificing the segmentation accuracy. Moreover, the authors replaced traditional convolutions with multi-scale 3D Res2Net modules and 3D residual hierarchical convolution (RHC). BRATS 2018 and BRATS 2020 were used to evaluate the performance of LHC-Net in this study, which was exhibited by the DSC (76.38% for ET, 90.01% for WT, and 83.32% for TC) with fewer parameters and fewer floating-point operations (FLOPs) than standard 3D U-Net models. Similarly, Zhang et al. [63] introduced HMNet, a hierarchical multi-scale network incorporating lightweight components such as 3D shuffle blocks and conditional channel weighting to balance computational cost. In this study, HMNet combined a high-resolution branch for spatial feature retention and multi-resolution branches to adapt to tumors of various shapes and sizes. The methodology integrates lightweight components, such as a 3D shuffle block and lightweight conditional channel (LCC) weighting block. The authors evaluated HMNet using the BRATS 2018, 2019, and 2020 datasets and achieved DSC values of 0.781, 0.901, and 0.823 for the ET, WT, and TC regions, respectively. While both networks effectively balance accuracy, HMNet enhanced ET region segmentation, whereas LHC-Net slightly outperforms HMNet in segmenting the TC region.

E. HYBRID FRAMEWORK

Different studies have presented hybrid frameworks with brain tumor segmentation and classification to support decision-making and enhance patient treatment by providing more details about the tumor for specialists and doctors. Additionally, Zaitoon and Syed [57] and Dang et al. [76] explored segmentation-enhanced deep learning frameworks by employing U-Net variants for 3D segmentation before tumor classification. Zaitoon and Syed [57] introduced RU-Net2+, leveraging skip connections, residuals, and recurrent units to refine segmentation accuracy, achieving 98.39% for HGG and 99.1% for LGG. The extracted tumor regions were then classified using a deep brain tumor CNN (DBT-CNN), reaching 99.51% and 99.28% accuracy for HGG and LGG, respectively. However, the study's dependence on the BRATS dataset constrains the model adaptability to real-world clinical cases, whereas complex DL model interpretability remains challenging. Dang et al. [76] adopted a multi-stage approach, utilizing 3D U-Net for segmentation, followed by VGG and GoogleNet classification on cropped background and non-background MRIs. Their approach achieved Dice scores of 0.91 (WT), 0.72 (TC), and 0.82 (ET) with a classification accuracy of 97.44%. However, a drawback of this study is the exclusion of images during the classification phase owing to the removal of non-tumor areas within the segmented regions. Zhuge et al. [43] developed classification frameworks that integrated 3D CNNs and segmentation-aided tumor classification to improve glioma grading. The authors designed a fully automated glioma classification pipeline that first segmented tumor regions using a 3D U-Net, followed by classification using a 3D ConvNet. Compared to the 2D Mask R-CNN baseline, which employed a ResNet backbone and achieved 96.3% accuracy, the 3DConvNet model resulted in marginally enhanced outcomes with 97.1% accuracy because of its focus on volumetric spatial information. However, this study acknowledges dataset imbalance challenges and potential generalization issues, particularly in defining LGG grades II and III. In conclusion, the literature shows that researchers have moderate interest in adopting 3D CNN models for brain tumor analysis. Notably, significant 3D CNN-based model enhancement has occurred from mid-2019 to 2024. This transition necessitated modifications in the data preprocessing and augmentation techniques to effectively handle 3D and volumetric data. Table 5 provides a comparative analysis of assorted studies and summarizes a careful analysis of the techniques used in the included studies, including the datasets utilized, CNN architecture implemented, primary characteristics of the studies, and reports the best performance results achieved.

VII. DISCUSSION

This study highlights the application of 3D CNN-based DL in automated brain tumor diagnosis based on multimodal MRI data. Through a thorough analysis of 32 articles, the review has explored different automated architectures of 3D CNN for brain tumor segmentation and classification between 2019 and 2024. This review has tracked the development of brain tumor analysis, focusing on segmentation and classification. The investigation found significant advancements achieved through 3D CNNs in volumetric feature extraction while underscoring persistent challenges limiting real-world integration. The study exhibited that the 3D CNN was widely used in the segmentation process in 24 studies (75%), with only five studies (16%) in the classification process and three studies (9%) having used both processes, as shown in Fig. 3.

To strengthen the contextual value of the included studies, a citation-based analysis was performed to identify the most prominent works within the reviewed literature. Among the 32 included studies, the most cited work was by Mzoughi [61], which proposed a multi-scale 3D CNN combined with small convolutional layers, accumulating around 347 citations. In contrast, Wang [55] had approximately 273 citations and developed triple-cascaded networks that utilize a 2.5D approach for hierarchically segmenting tumor regions. The study by Zhuge [43], which developed a fully automated framework by segmenting tumor regions and classifying tumor types, garnered around 169 citations. Montaha [60] developed TD-CNN-LSTM for brain tumor classification, which ranked among the top performers and achieved 128 citations. While Ali [75] evaluated a 3D U-Net with an autoencoder structure for the segmentation tumors and received approximately 127 citations. These highly referenced studies have influenced the development of 3D CNN models in brain tumor research and remain essential references for current progress in the domain.

The included studies demonstrated the potential of 3D CNN-based structures to extract spatial relations and features in volumetric MRI data more effectively than 2D models like transformers in capturing inter-slice information, which helps understand the structure and context of tumors within 3D MRI. In addition, this review shows that integrating multisequence data with 3D data to extract complementary information improves segmentation accuracy. In classification, the studies have highlighted the effectiveness of 3D CNNs using volumetric data to classify subtypes of the glioma tumor, as illustrated in Fig. 7. Combining spatial context with temporal or semantic feature models resulted in classification accuracy exceeding 98% in hybrid frameworks such as TD-CNN-LSTM [60] and models incorporating fuzzy logic [59]. CNN-based 3D U-Net architecture and their configurations have shown superiority over GANs and 2D techniques for segmentation of brain tumors, especially within WT and TC regions, as shown in Fig. 8.

The GANs have been leveraged for data increasing and semi-supervised training to offer utility in mitigating data scarcity but do not typically serve as a standalone segmentation process. While the 3D CNNs offer the most balanced approach by integrating spatial continuity and contextual awareness. In general, the DSC distributions show a gradual improvement in all regions of tumor between 2019 and 2024, as depicted in Fig. 9. Different 3D approaches achieved

TABLE 5. A comprehensive analysis of the included studies that concentrate on 3D CNN models for brain tumor segmentation and classification using multimodal MRI scans.

Study Details	Datasets	Architectures	Augmentation	Preprocessing	Computational Cost	Best Results (%)
Classification Frameworks						
Rani et al. [42]	BRATS 2019, Figshare, Kaggle, and Self-collected dataset.	3D AlexNet with CLAP	N/A	Adaptive median filter	GPU (32 GB RAM), 65 epochs, Batch size: 128, LR: 0.001, Dropout: 0.2, early stopping, ~3700–3900 s inference time.	Acc: 96.91
Wu et al. [58]	BRATS 2018 and 2019.	Attention mechanism in AGGN	Rotation, Translation, and Clipping	Cropping	NVIDIA GTX 2080Ti GPU, 100 epochs, batch size 32, 16.37M parameters, 24.8 GFLOPs.	Acc: 96 Pre: 99.8 Sen: 95 Spe: 99.5 F1-score: 94.5
Molaei et al. [59]	BRATS 2012-2018 (145 patients and 1000 images).	Fuzzy deep CNN (FDCNet)	N/A	Denoising, and contrast enhancement	MATLAB 2015b on a 7-core CPU 3.6 GHz, 6 GB RAM, Windows 8; Error probability converged within ~200 seconds.	Acc: 98.68 Sen: 94.87
Montaha et al. [60]	BRATS 2018, 2019, and 2020.	TimeDistributed-CNN-LSTM	N/A	Min-max normalization, cropping, and resizing	128×128×32 input size, batch size 16, 150 epochs, optimized for GPU constraints, model complexity ≈116M operations.	Acc: 98.90 Spe: 99.15 Pre: 98.95 Sen: 98.78 F1-score: 98.8
Mzoughi et al. [61]	BRATS 2018	Deep multi-scale 3D CNN	Flipping	Min-max normalization, resizing, and contrast enhancement	GTX 1080 Ti (11 GB), Intel i7, 19.5 GB RAM.	Acc: 96.49
Segmentation Frameworks						
Zhou et al. [54]	BRATS 2017	3D U-Net-based multi-stage framework	N/A	N4ITK	Single GPU (Quadro P5000, 16GB).	WT: 89.4 TC: 81.6 ET: 73
Wang et al. [55]	BRATS 2017 and 2018.	Triple-cascaded framework by anisotropic 2.5D CNN	Rotation, flipping, and scaling	Z-score normalization	NVIDIA TITAN X GPU (12 GB).	WT: 0.902 ± 0.056 TC: 0.858 ± 0.139 ET: 0.797 ± 0.229
Ilyas et al. [56]	BRATS 2017 and 2018.	Multi-dilated attention network (Hybrid-DANet)	Cropping	Cropping	NVIDIA GTX 3070 (8 GB) GPU using PyTorch.	Brats 2017: WT: 89.2 TC: 76.4 ET: 68 Brats 2018: WT: 87.6 TC: 74.6 ET: 63.9
Li et al. [62]	BRATS 2018 and 2019.	Modified 3D U-Net	N/A	Z-score normalization and cropping	AMD 5950X CPU, 64GB RAM, NVIDIA RTX 3090 (24GB); 100 epochs, batch size 4, PyTorch.	WT: 77.8 TC: 87.5 ET: 90.3
Zhang et al. [63]	BRATS 2018, 2019, and 2020.	Lightweight components with HMNet	Rotation and flipping	Z-score normalization	Trained on Intel i9-9900X, NVIDIA GTX 2080 Ti (11GB), 0.80M Params, 129.4 GFLOPs, batch size 2, 500 epochs, PyTorch.	WT: 90.1 TC: 82.3 ET: 78.1

TABLE 5. *(Continued.)* **A comprehensive analysis of the included studies that concentrate on 3D CNN models for brain tumor segmentation and classification using multimodal MRI scans.**

Zia et al. [64]	BRATS 2018, 2019, and 2020.	Context-aware 3D ARDUNet (RDBs + ARDBs)	N/A	Min-max normalization and cropping	Intel i7 CPU, 16 GB RAM, NVIDIA GTX 1060 (6 GB), trained >1100 epochs using Keras (TensorFlow backend).	WT: 92.75 TC: 92.79 ET: 93.65
Chen et al. [65]	BRATS 2018 and 2020.	MTDC-Net-CNN (SPC + MTAU)	Flipping and clipping	Min-max normalization and cropping	Intel Xeon Gold 6240 @ 2.60 GHz, 376 GB RAM, NVIDIA Tesla V100; Training: 2 s/batch, Testing: 1 s/batch, Segmentation: 3 s/image.	WT: 89.2 TC: 83.2 ET: 80.9
Nehru and Prabhu [66]	BRATS 2018 and 2019.	DSDU-Net	Rotation	Min-max normalization, N4ITK, unsharp masking, and histogram equalization	N/A	WT: 91 TC: 90 ET: 86
Zhou et al. [67]	BRATS 2018	Tri-attention fusion network (U-Net-based approach)	N/A	Cropping and N4ITK	40h training on NVIDIA Quadro P5000 (16GB), 3D input (128×128×128), 6-layer model, 8 initial filter.	WT: 88.7 TC: 79.6 ET: 75
Ahmad et al. [68]	BRATS 2018, 2019, and 2020.	MH-UNet	Rotation and flipping	Cropping	Trained for 300 epochs using Adam optimizer on 128^3 patches with batch size 1, implemented in Keras on a single 32 GB GPU.	WT: 77.8 TC: 87.5 ET: 90.3
Ullah et al. [69]	BRATS 2018	3D U-Net	N/A	Nyul intensity, z-score normalization, N4ITK, AHE, Gibbs ringing artifact removal	NVIDIA Titan X Pascal GPU, 3D U-Net trained for 80 epochs with mini-batch size of 2 and patch size 80×80×80.	WT: 91 TC: 86 ET: 70
Wang et al. [70]	BRATS 2018 and 2020.	LHC-Net (3D Res2Net module + 3D RHC module)	Flipping and shifting	Z-score normalization, cropping, and N4ITK	1.65M params; 35.58 GFLOPs, NVIDIA RTX 3090 GPU for 450 epochs (batch size 4), Implemented in PyTorch 1.8.	WT: 90 TC: 83 ET: 76
Lachinov et al. [71]	BRATS 2018	Cascaded 3D U-Net	Gaussian noise and flipping	Z-score normalization	4× GTX 1080 Ti GPUs, 500 epochs, batch size 4, trained with data parallelism using MXNet.	WT: 87.8 TC: 79.5 ET: 72
Yin et al. [72]	BRATS 2019	ConvNeXt blocks with WD-UNeXt	N/A	Z-score normalization, cropping, and N4ITK	NVIDIA RTX 3090 (24GB, PyTorch 1.10, CUDA 11.0), 300 epochs with checkpointing to reduce memory, 29.6M parameters; 39.6 GFLOPs computational complexity.	WT: 93.4 TC: 91.1 ET: 85.1
Ali et al. [73]	BRATS 2019	ML-KCNN (CCA + CRF)	Rotation and Gaussian noise	Z-score normalization	Google Colab with an NVIDIA K80 GPU. The test prediction time ranged from 15–20 seconds (ML-KCNN) and increased to 30–40 seconds with post-processing.	WT: 90 TC: 83 ET: 74
Tie et al. [74]	BRATS 2019	3D U-Net (Dense + Res blocks)	Rotation	Z-score normalization	PyTorch on NVIDIA GTX1070 GPU (8GB), batch size 1, 260 epochs, patch size 4×96×96×96. Adam optimizer, learning rate 1e-4 (decayed).	WT: 90.1 TC: 81.5 ET: 76.6
Ali et al. [75]	BRATS 2019	3D CNN and 3D U-Net	Rotation, flipping, and cropping	Z-score normalization and cropping	NVIDIA Geforce GTX1070 GPU with 8 GB memory, involved 3D CNN and 3D U-Net with 128^3 patch sizes and batch sizes of 2, trained for 150 epochs.	WT: 90.6 TC: 84.6 ET: 75

TABLE 5. *(Continued.)* **A comprehensive analysis of the included studies that concentrate on 3D CNN models for brain tumor segmentation and classification using multimodal MRI scans.**

Aboussaleh et al. [77]	BRATS 2020	3D UV-NetR+ (U-Net + V-Net)	N/A	Z-score normalization and resizing	NVIDIA Quadro P6000 (24GB GPU), TensorFlow, 200 epochs, batch size 2, training time: 4.6 h, 11.7M parameters.	WT: 91.95 TC: 82.80 ET: 81.70
Rutoh et al. [78]	BRATS 2020	3D GAID-U-Net	N/A	Z-score normalization and resizing	27M params, 72.68 GFLOPs; 1658 mins training on A100 GPU (40GB, Colab Pro+); 3.5s/test case.	WT: 88.7 TC: 86.2 ET: 85.7
Khan et al. [79]	BRATS 2020	HA-RUnet	N/A	Z-score normalization and resizing	NVIDIA Titan 1080 Ti (12GB), 100 epochs, batch size 1, ~5.2 min/epoch (total ~8.7 hours); 13.2M parameters.	WT: 86.7 TC: 81.3 ET: 78.7
Nazir et al. [80]	BRATS 2020	3D KCFP (3D FSC + 3D CCA)	Flipping	Z-score normalization and Gaussian blurring	Intel Core M-5Y10c CPU @ 1.80 GHz (8 GB RAM), NVIDIA Titan X Pascal GPU (3584 CUDA cores) with ~3M parameters, 400 epochs, patch size 128×128×128. Average inference time ≈ 5s/patient.	WT: 90 TC: 84 ET: 80
Magadza and Viriri [81]	BRATS 2020	partial depth-wise 3D U-Net	Gaussian noise and translation	N/A	NVIDIA Tesla V100 (16GB), batch size 2, 100 epochs; ~26MB model size ~70% fewer parameters vs. baseline.	WT: 87.3 TC: 85 ET: 76.6
Shaukat et al. [82]	BRATS 2020	3D U-Net (cloud-based model)	N/A	Z-score normalization and cropping	Azure NC6 VM with 1× Tesla K80 GPU (24 GB), 56 GiB RAM, 6 vCPUs; batch size 8, 100 epochs; $1.321/hour; training time not reported.	DSC: ~ 95
Liu and Cheng [83]	BRATS 2021	3D BCS-CNN	Flipping, shifting, scaling, deformation, and Gaussian noise	N4ITK	4× Tesla V100 GPUs, Dual Intel Xeon Gold 5218 CPUs, 256 GB RAM, Training: 300 epochs (~21 min/epoch); Batch size: 4, Inference time <3s.	WT: 83.9 TC: 88 ET: 86.7
Hybrid Frameworks						
Zhuge et al. [43]	BRATS 2018 and TCGA-LGG.	3D U-Net for segmentation and 3D ConvNet	Rotation, flipping, and scaling	Z-score normalization, N4ITK	DELL T7910, 2×NVIDIA Titan Xp (12 GB), 3D segmentation: ~10.5 h (300 epochs), 2D grading: ~4 h (60k iters), 3D grading: ~2 h (10k iters), Testing: 50 s (3D), 1 s (2D), 0.25 s (3D).	Acc: 97.1 Sen: 94.7 Spe: 96.8
Zaitoon and Syed [57]	BRATS 2017, 2018, 2019, and 2020.	RRU-Net2+for segmentation and 3D DBT-CNN for classification	N/A	CNMF	Google Colab with GPU acceleration (57× speed-up over CPU). Trained for 10 epochs with batch size 256 using TensorFlow 2.1.0 and Keras 2.3.1.	Acc: HGG: 99.51 LGG: 99.28. DSC: HGG: ~98.5 LGG: ~98.8
Dang et al. [76]	BRATS 2019	3D U-Net for segmentation and 3D VGG and GoogleNet for classification	Rotation	Gamma correction, WSO	Batch size: 1 (segmentation) due to memory limits, model simplified for 3D input.	Acc: 97.44 DSC: WT: 91 TC: 72 ET: 82

DSC frequently exceeding 90% in WT regions, including context-aware 3D ARDUNet (e.g., RDBs and ARDBs) [64], depth-wise separable dense U-NET [66], ConvNeXt blocks with the WD-UNeXt model [72], hybrid U-Net-VeNet models [77], and CNN-based 3D U-Net [69], [74], and [75]. However, the ET region often achieved lower DSC scores, between 70% and 85%, due to its smaller size and variability.

The success of CNN architectures in brain tumor diagnosis is deeply dependent on the integrity of the data. The

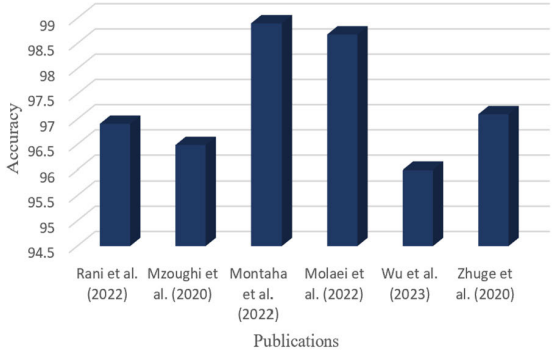


FIGURE 7. Comparison of classification accuracy across 3D CNN-based models.

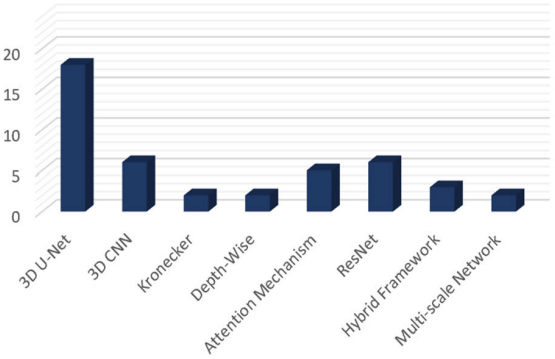


FIGURE 8. Prevalence of segmentation architectures (U-Net, CNN, ResNet, etc.) in included studies.

CNNs require a complete and consistent dataset to make accurate predictions and diagnoses. Data preprocessing techniques were essential for improving intensity uniformity and reducing noise to enhance the performance of CNNs; however, they introduced inconsistencies across studies due to varying implementations. Therefore, any variation in data quality can lead to CNN models making unreliable predictions. In addition, CNN models require large datasets to train complex patterns for more comprehensive learning of brain tumor. This review reveals that the CNN models have specific data-related constraints, such as the class imbalance challenge. The unequal spread of tumor and non-tumor in MRI and the uneven representation of tumor subtypes cause the class imbalance challenge, which impacts the classification and segmentation performance of brain tumor analysis. This issue could cause bias since the model is more proficient in well-represented data, while it performs poorly in underrepresented data. Therefore, it is crucial to create a balanced dataset. Ali et al. [73] used rotation and Gaussian noise with a generalized dice loss (GDL) function to address the class imbalance problem by assigning the weight of each class inversely to the square of label distributions. In [77], [80], and [81], their models did not achieve adequate dice scores for the ET because the tumor classes in the dataset were not evenly distributed. During training, data augmentation techniques reset and increase the representation of less prevalent data. This step is crucial to counter the issues of small data samples and class imbalances, ensuring that the CNN models can learn with equal precision via varied samples.

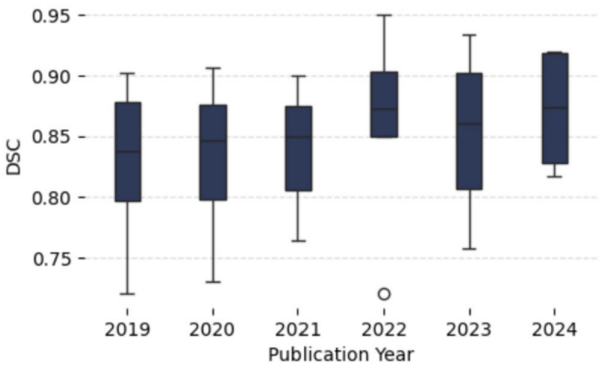


FIGURE 9. DSC distribution of 3D CNN-based models segmentation results from 2019 to 2024.

In MRI modalities analysis using 3D CNN, the high computational cost of the networks constitutes a significant obstacle to the clinical application due to the high resource demands associated with processing volumetric data for brain tumor analysis. These models often require high GPU memory and long training times, especially when employing multi-branch architectures, such as multi-stage frameworks [54], [76], cascaded architectures [55], [71] and attention-guided networks [67], [78]. Yin et al. [72] required high-end GPUs like the NVIDIA RTX 3090 (24GB) and extensive RAM for training, whereas lightweight models like LHC-Net achieved reasonable performance using fewer parameters and FLOPs on RTX 3090 with only 1.65M parameters. Similarly, Zhang et al. [63] used the GTX 2080 Ti with only 0.8M parameters, reducing hardware requirements significantly. While these models maintain high performance in WT and TC regions, they tend to exhibit slightly lower accuracy in ET segmentation, reflecting the trade-off between efficiency and granularity of tumor boundary delineation. These comparisons indicate that while high-performing models often demand expensive hardware, emerging lightweight architectures can offer a practical compromise for deployment in resource-limited settings.

3D CNN models for brain tumor analysis are often intense learning models, as deeper and more developed frameworks are needed to extract the substantial spatial features and heterogeneous nature of brain tumors. This increase in architectural depth and parameters is called model complexity, which can lead to overfitting and difficulty in optimizing the network. The complex architecture of many proposed models hinders scalability, as not every region or health agency is provided with high-specification computers and qualified resources. Approaches like residual and recurrent networks [57], modified 3D U-Nets with multi-threading dilated networks [65] or hybrid convolution-transformer designs [77], [83] increase the depth of networks, which enhances feature extraction and accuracy rate but increases hyperparameter tuning and overfitting. Shaukat et al. [82] and Nazir et al. [80] highlighted that deeper models improved performance but led to increased non-linearity of networks

and training instability. However, there is still room for performance improvement to balance efficiency and accuracy in brain tumor detection.

Despite the increasing accuracy of 3D CNN-based models, their lack of interpretability raises a challenge for clinical acceptance, as these models often operate as opaque systems whose decision-making processes are not easily understood by humans. At the same time, most reviewed studies focused on performance metrics, and few incorporated explainable AI (XAI) techniques to interpret model outputs. Important methods like saliency maps, gradient-weighted class activation mapping (Grad-CAM), and SHapley additive exPlanations (SHAP) are commonly used in 2D CNN applications to show which parts of an image are important or to measure how much unique features contribute. These techniques can show which parts of an image, like tumor regions, affect the decisions made during classification or segmentation, helping us understand the results better. However, their application to 3D CNNs remains limited due to computational complexity, lack of native support in most toolkits, and the challenge of visualizing volumetric explanations. Despite these constraints, emerging research has demonstrated that adapted versions of Grad-CAM and Deep SHAP can be extended to 3D models, enabling clinicians to interpret predictions over spatial volumes.

The deployment of 3D CNN models into real-world clinical workflows remains limited. A major obstacle is the lack of external validation on real-world and multi-institutional data with diverse acquisition procedures. As mentioned earlier, challenges like imbalanced data, computational cost, model complexity, and interpretability limit the generalization of the models and real-world clinical acceptance. In addition, most studies depend on publicly available datasets for training and testing processes, which may not consider the variability of clinical settings. Future works in brain tumor classification and segmentation, leveraging 3D CNN architectures, could consider several directions to facilitate clinical adoption, address the identified challenges, and enhance the performance models. These potential directions include:

- Designing Interpretable AI Models: Future research should focus on integrating XAI methods in 3D CNNs to improve clinical trust. Methods like 3D-adapted Grad-CAM and Deep SHAP could be enhanced to reveal how various areas of tumors in 3D MRI scans influence the model's decisions, assisting radiologists and doctors in verifying those decisions for diagnosis and treatment planning.
- Developing Computationally Efficient Architectures: Lightweight 3D CNN architectures that decrease GPU memory cost and reduce training time without affecting segmentation or classification accuracy are needed. Approaches like depth-wise separable convolutions and 3D-adapted transformers could help manage the resource limitations in clinical environments.
- Enhancing Data Augmentation: In small tumor regions like ET, research should continue developing robust augmentation pipelines, such as GANs or synthetic data generation that mimic the statistical properties and patterns of real-world data, to improve learning efficacy across all tumor classes and regions.
- Standardizing Preprocessing Pipelines: Variability in preprocessing techniques and evaluation metrics over studies complicates comparison and reproducibility. Developing standardized procedures for bias field correction, normalization, and performance benchmarking would support fair evaluation of research findings.
- Personalized DL Models: Future works could explore the development of personalized DL models that consider the unique characteristics of each patient, such as genetic information, lifestyle factors, and comorbidities. Personalized models could lead to more tailored interventions, which improve patient outcomes.
- Cross-disciplinary Collaborations: Collaboration with radiologists during model design will further bridge the gap between academics and doctors. These collaborations can ensure that AI technologies are developed with a profound understanding of clinical needs.

VIII. CONCLUSION

This systematic literature review introduces a comprehensive technical analysis of 3D CNN-based DL architectures for brain tumor diagnosis based on 3D multimodal MRI data. Adopting 3D CNN-based models, including 3D U-Net, hybrid frameworks, Kronecker techniques, attention mechanisms, and multi-scale networks, has significantly enhanced the performance of segmentation and classification of brain tumors and early disease detection. These techniques play a role in addressing the complex nature of brain tumor within the MRI scan, which typically contends with various inconsistent data. 3D CNNs' integration into CAD systems enhances the accuracy and efficiency of detecting brain tumor regions. Although the benefits of 3D CNN for volumetric data in brain tumor diagnosis are clear, continued research and technological improvement are important to expand its effectiveness and ensure reliable integration into daily clinical settings.

To enhance the integrity of the findings and guide future research, it is crucial to address the limitations identified in this review. A primary concern is the generalizability of the results, which are primarily derived from high-impact journals. This selection may overlook evolving studies from lesser-known platforms. Future research should include a wider variety of sources, thereby enriching the diversity and inclusivity of the research examined.

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